



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Faculty of Engineering, Built Environment and
Information Technology

EAI 320

INTELLIGENT SYSTEMS

PRACTICAL 4 GUIDE V1.1

UPDATED ON 29 APRIL 2025

Concept: Prof. Warren P. du Plessis

Guide written by: Llewellyn Strydom

Revised by: Michael Teles, Steven Pinto, Liebet Joubert, David Grünewald, and Rynhardt Luies.

Due date: 26 May 2024 @ 23:59

I. SCENARIO

The goal of this assignment is to implement a naïve Bayes classifier. The naïve Bayes classifier is an algorithm that solves classification problems by making a simple assumption: that all features in the input vector $\mathbf{f}$ are conditionally independent. Although this assumption does not always hold, naïve Bayes classifiers have been widely used and have shown to perform well on a variety of tasks.

The naïve Bayes classifier is based on Bayes' law, which can be written as

$$p(C_k|\mathbf{f}) = \frac{p(\mathbf{f}|C_k)p(C_k)}{p(\mathbf{f})}, \quad (1)$$

where $p(C_k|\mathbf{f})$ is the probability that an object belongs to a certain class C_k given a feature vector $\mathbf{f}$. Since the denominator of Bayes' law acts simply as a normalisation term, only the numerator need be considered when implementing a naïve Bayes classifier. The numerator term can be rewritten using the product rule of probability [1] as

$$\begin{aligned} p(\mathbf{f}|C_k)p(C_k) &= p(f_1, \dots, f_n, C_k) \\ &= p(f_1|f_2, \dots, f_n, C_k) p(f_2, \dots, f_n, C_k) \\ &= p(f_1|f_2, \dots, f_n, C_k) p(f_2|f_3, \dots, f_n, C_k) p(f_3, \dots, f_n, C_k) \\ &= \dots \\ &= p(f_1|f_2, \dots, f_n, C_k) p(f_2|f_3, \dots, f_n, C_k) \dots p(f_{n-1}|f_n, C_k) p(f_n|C_k) p(C_k). \end{aligned} \quad (2)$$

If the elements from the feature vector are considered to be conditionally independent then Eq. 1 simplifies to

$$p(C_k|\mathbf{f}) \propto p(C_k) \prod_{i=1}^F p(f_i|C_k), \quad (3)$$

where F is the number of features.

The naïve Bayes classifier implements the following simple decision rule to select the class with the highest posterior probability $\hat{C}$ for classification:

$$\hat{C} = \underset{k \in \{1, \dots, K\}}{\operatorname{argmax}} p(C_k) \prod_{i=1}^F p(f_i|C_k). \quad (4)$$

For more information on naïve Bayesian classifiers please refer to the prescribed textbook [2].

II. INSTRUCTIONS

The task is to implement an **rock-paper-scissors (RPS)** agent that implements a version of the naïve Bayes classifier that is able to learn in an online fashion. The agent will thus be expected to learn the strategy of an arbitrary opposing agent by recording the history of moves played during a game and calculating the relevant probabilities that are required to implement a naïve Bayes classifier. The agent must not be pre-trained. It must be able to learn the strategy of its opponent online while playing against it.

It is recommended to use vector operations in MATLAB to ensure efficient execution of your code.

III. SUBMISSION REQUIREMENTS

Students are not required to submit a technical report for this **INDIVIDUAL** assignment. As there is not technical report for this practical, there is a strong requirement that the submitted code be well-commented and easy to understand. A mark of zero will be awarded if the submitted code file does not run, produces errors, or does not follow the instructions.

Add a brief discussion and explanation of the results and observations as a comment at the end of your code.

A. Code Instructions

- All code must be commented to a point where the implementation of the underlying algorithm can be determined.
- Your name and student number must be included as comments at the top of the submitted code file.
- No late assignments will be accepted. No excuses for late submission will be accepted.
- Each student must do their own work. Academic dishonesty is unacceptable and cases will be reported to the university Legal Office for suspension.
- All information from other sources must be clearly identified and referenced.
- Any attempt to interfere with the operation of the framework, excluding the test code on the MATLAB Grader, used (e.g. to modify the scores of agents) will be regarded as academic dishonesty.

B. Submission

- Code should be submitted to the MATLAB Grader via the following link on the EAI 320 **ClickUP** page; Practicals → Practical 4 → Naïve Bayes Classifier .
- Submit your code on TurnItIn with the EECE cover page. Name the file `eai320_{practical number}_{student number}.pdf`. Do not email code to the lecturer or **Assistant Lecturer (AL)** as emailed code will be considered not to have been submitted.
- Submit your code to the **Judicator automated marking system (AMS)** with the name `eai320_{practical number}_{student number}.m`.

Only assignments submitted via ClickUP will be accepted. Do not email code to the lecturer or **AL** as emailed code will be ignored.

Late assignments will not be accepted! Accepting late assignments is extremely unfair to those students who submit their work timeously because their tardy colleagues are effectively given additional time to complete the same work. Students are advised to submit the day before the deadline to avoid inevitable problems with ClickUP, internet connections, unsynchronised clocks, load shedding, hard-drive failure, computer theft, etc.. Students who choose to submit close to the deadline accept the risk associated with their actions, and no excuses for late submissions will be accepted.

Students will be allowed to submit updated copies of their assignments until the deadline, so there will be no excuse for submitting late. **Rather be marked on an incomplete early version of your assignment than fail to submit anything.**

C. Academic Dishonesty

Academic dishonesty is completely unacceptable. Students should thus familiarise themselves with the University of Pretoria's rules on academic dishonesty summarised in the study guide and the university's rules. Students found guilty of academic dishonesty will be reported to the Legal Office of the University of Pretoria for suspension.

Students should clearly indicate material from other sources and provide complete references to those sources. Examples of commonly-used sources include the textbook [2] and this document [3]. Note that this does not mean that students may reuse code and/or information found in books, on the internet, or in other sources as students are required to complete the tasks themselves.¹

REFERENCES

- [1] C. M. Bishop, *Pattern Recognition and Machine Learning*. Springer-Verlag, 2006.
- [2] S. J. Russell and P. Norvig, *Artificial intelligence: A modern approach*, 3rd ed. Prentice Hall, 2010.
- [3] L. Strydom, *EAI 320 – Practical 6 Guide*, University of Pretoria, 25 Apr. 2019.

¹The objective of all academic assignments is fundamentally that students learn by completing the assignments. Merely reusing code and/or information found elsewhere defeats this objective because a key part of the learning process is performing the tasks oneself.

APPENDIX A MARKING RUBRIC

Question 1: Instructions and Formatting	5
Instructions	
– Correct file naming convention	1.25
– Student information comment	1.25
Formatting	
– Descriptive commenting	1.25
– Descriptive variable naming convention	1.25
Question 2: Discussion	10
Clear description of fundamental principles of the naïve Bayes classifier (in the comments).	10
Question 3: Simple agent	5
Win 90% of rounds and 90% of matches.	5
Question 4: Smart agent	10
Win 60% of rounds and 90% of matches.	10
Question 5: Complex agent	10
Win 30% of rounds and 70% of matches.	10
Total	40

APPENDIX B ABBREVIATIONS

AL	Assistant Lecturer
AMS	automated marking system
RPS	rock-paper-scissors